

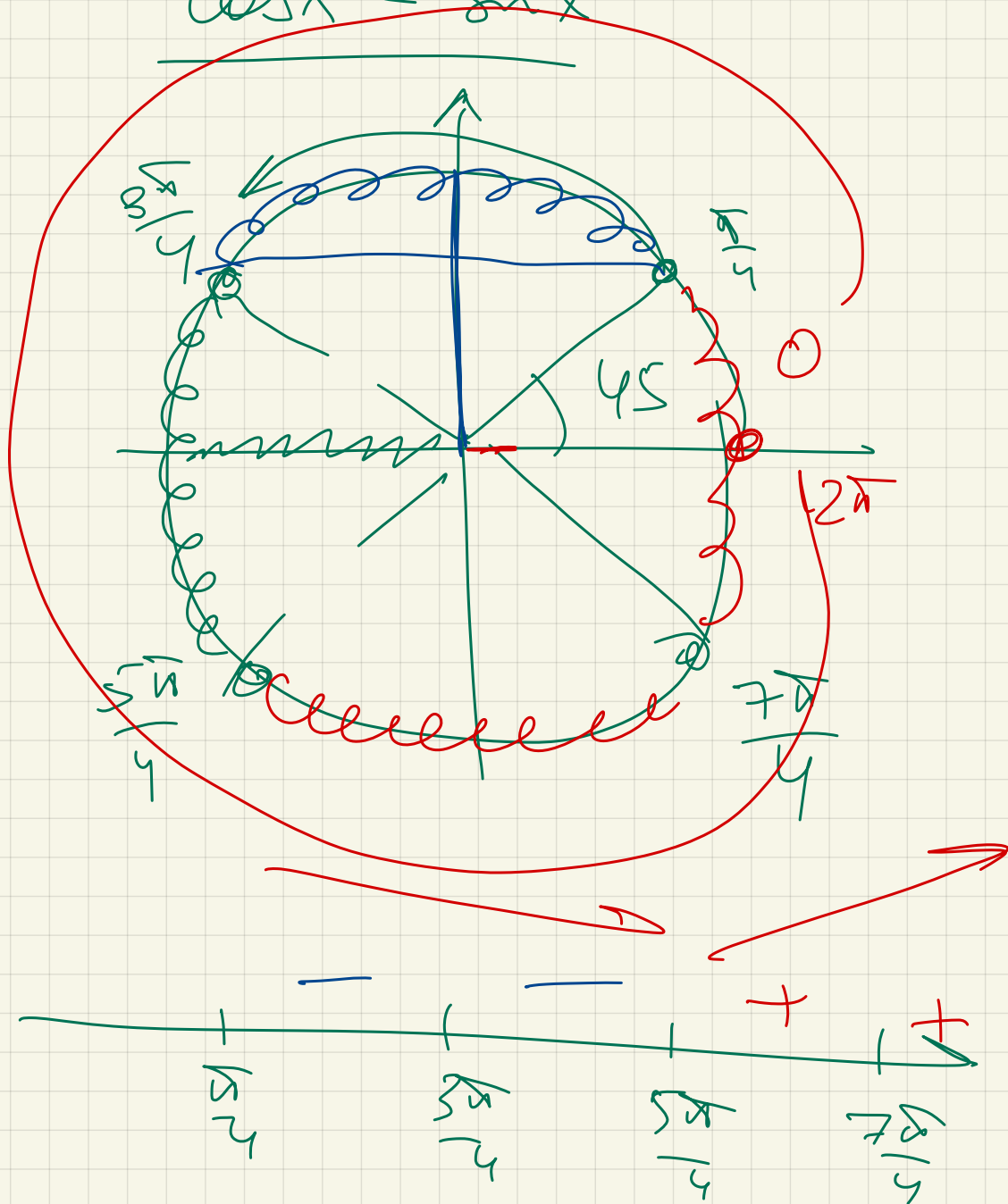


$$e^x (-\sin x) + e^x \cos x$$

$$e^x (-\sin x + \cos x) = 0$$

①

$$\cos x = \sin x$$



$$f(x) = x + \frac{1}{x}$$

$$f'(x) = 1 + (x^{-1})' = 1 - \frac{1}{x^2}$$

$$= -1 \cdot x^{-1-1} = -\frac{1}{x^2}$$

$$\frac{x^2 - 1}{x^2}$$

$$f(x) = x^2 e^{-x}$$

$$f'(x) = 2x e^{-x} + (-1) e^{-x} \cdot x^2 =$$

$$x e^{-x} (2 - x) = 0$$

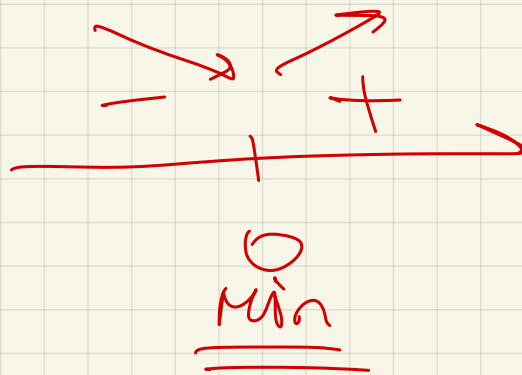
$x = 0$ $x = 2$

$\begin{array}{c} \nearrow \quad \nearrow \quad \nearrow \\ \text{---} \quad \text{---} \quad \text{---} \\ \text{---} \quad \text{---} \quad \text{---} \end{array}$

$\text{min} \quad \text{max}$

$$x(2-x) = -1(2-(-1)) = -1(2+1) = -1 \cdot 3 = -3$$

$$f(x) = \frac{2x}{x^2+1}$$



$$f(x, y) = 2x^2 + 2xy + y^2 - 4x - 2y \quad \exists$$

$$f'_x = 0$$

$$f'_y = 0$$

$$f'_x = 4x + 2y - 4$$

$$f'_y = 2x + 2y - 2$$

$$\begin{cases} 4x + 2y - 4 = 0 \\ 2x + 2y - 2 = 0 \end{cases}$$

$$\begin{cases} 2x + y - 2 = 0 \\ x + y - 1 = 0 \end{cases}$$

$$y = 0$$

$$x = 1$$

$$(1, 0)$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$$

$$(f'_y)'_x$$

$$f''_{xx} = A$$

$$f''_{xy} = B$$

$$f''_{yy} = C$$

$$D = AC - B^2$$

$$f''_{xx} = 4$$

$$f''_{yy} = 2$$

$$f''_{xy} = 2$$

$$D = 4 \cdot 2 - 4 = 8 - 4 = 4 > 0$$

$\Rightarrow$ min

$$f(x; y) = x^3 + y^3 - 3xy$$

(1:1) экстремум?

$$f'_x = 3x^2 - 3y$$

$$f'_y = 3y^2 - 3x$$

$$\begin{cases} f'_x = 0 \\ f'_y = 0 \end{cases}$$

$\Rightarrow$ need
y.cn.
экс

✓

$$f''_{xx} = 6x = A$$

$$f''_{xy} = -3 = B$$

$$f''_{yy} = 6y = C$$

$$AC - B^2$$

$$6x \cdot 6y - (-3)^2$$

$$36xy - 9 = D$$

$$D(1:1) \geq 0 \quad \textcircled{V} \rightarrow A > 0$$

min

$$f(x,y) = 4xy - x^4 - y^4$$

$$\left. \begin{array}{l} (0;0) \\ (1;1) \\ (-1;-1) \\ (1;-1) \\ (-1;1) \end{array} \right\} \circ$$

$$f'_x = 4y - 4x^3$$

$$f'_y = 4x - 4y^3$$

$$(0;0) +$$

$$(1;1) +$$

$$(-1;-1) +$$

$$\cancel{(1;-1)} \ominus$$

$$\cancel{(-1;1)} \ominus$$

$$f''_{xx} = -12x^2$$

$$f''_{xy} = 4$$

$$f''_{yy} = -12y^2$$

$$A = -x^2$$

$$\textcircled{1} \quad D = 144x^2y^2 - 16 < 0 \quad \textcircled{X}$$

$$(0;0) \quad A = -12x^2 =$$

$$(1;1)$$

$$\begin{array}{l} A > 0 \\ A < 0 \end{array}$$

$$\max \quad \textcircled{V}$$

$(-1:-1)$

$D > 0$

$A < 0$

max $\bigcirc \checkmark$